IB MATH AA SL

Lesson 42

Definite Integrals & Area

Calculus · 60-minute lesson

Syllabus SL 5.5 · Property of Lynda Badmus Education

Do Now — recall

ON YOUR WHITEBOARD

1. Find $\int 2x\,dx$.
2. Evaluate x^2 at $x = 3$ then at $x = 1$.
3. $3^2 - 1^2 = ?$

Answers (click to reveal)

1. $x^2 + C$
2. 9 and 1
3. 8

What we're learning today

LEARNING OBJECTIVES

1. **Evaluate** a definite integral.
2. **Apply** the limits (top minus bottom).
3. **Find** the area under a curve.
4. **Use** the GDC for definite integrals.

SUCCESS CRITERIA — I CAN...

- ✓ integrate, then substitute the limits
- ✓ compute F(b) − F(a)
- ✓ interpret a definite integral as area
- ✓ use the GDC to check the value

Definite integrals

WITH LIMITS

$$\int_a^b f(x)\ dx = F(b) - F(a)$$

Integrate
find F(x) (no + C needed here)

Substitute
top limit minus bottom limit

NO + C

The constant cancels in F(b) − F(a).

So
omit C for a definite integral.

Evaluate a definite integral

QUESTION

Evaluate $\int_1^3 2x\,dx$.

SOLUTION

Integrate: $F(x) = x^2$.

$$\int_1^3 2x\ dx = [x^2]_1^3$$

Top minus bottom ($3^2 - 1^2$):

$$= 9 - 1 = 8$$

Your turn — definite

ANSWER IN YOUR BOOK

1. $\int_0^2 2x\,dx$.

2. $\int_1^2 3x^2\,dx$.

3. $\int_0^1 4x\,dx$.

Answers (click to reveal)

1. $[x^2]_0^2 = 4$
2. $[x^3]_1^2 = 8-1 = 7$
3. $[2x^2]_0^1 = 2$

Area under a curve

AREA = INTEGRAL

$$Area = \int_a^b f(x)\ dx$$

Above the x-axis
the definite integral gives the area

GDC
use the definite-integral function to evaluate

CAREFUL

Below the axis

gives a negative — area is its absolute value.

Limits

a and b are the x-bounds of the region.

Area under a curve

QUESTION

Find the area between $y = x^2$, the x-axis, and the lines $x = 0$ and $x = 2$.

SOLUTION

Area = $\int_0^2 x^2\,dx$; integrate to $x^3/3$:

$$\left[\frac{x^3}{3}\right]_0^2$$

Top minus bottom (8/3 − 0):

$$= \frac{8}{3}$$

EXAM-STYLE QUESTION

Exam-style question

(a) Evaluate $\int_1^4 (2x + 1)\,dx$.   [3]

(b) Hence state the area between $y = 2x + 1$, the x-axis and $x = 1$, $x = 4$.   [1]

MARK SCHEME — reveal after students attempt (tutor only)

(a) $[x^2 + x]_1^4 = (16+4) - (1+1)$ (M1)(A1) $= 18$ (A1)

(b) area = 18 (square units) (A1)

Common mistakes

No + C
Drop the constant for a definite integral.

Order
It is F(b) − F(a): top limit first.

Negative area
A region below the x-axis integrates to a negative.

Square brackets
Write $[F(x)]_a^b$ to show the limits.

Mixed practice

QUESTIONS

1. $\int_0^3 2x\,dx$.
2. $\int_1^2 4x\,dx$.
3. Area under $y = x^2$ from 0 to 3.
4. $\int_0^1 6x^2\,dx$.

Answers (click to reveal)

1. $[x^2]_0^3 = 9$
2. $[2x^2]_1^2 = 8-2 = 6$
3. $[x^3/3]_0^3 = 9$
4. $[2x^3]_0^1 = 2$

Plenary & exit ticket

KEY FORMULAS

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

$$Area = \int_a^b f(x)\,dx$$

EXIT TICKET — before you leave

1. $\int_0^2 2x\,dx$.
2. Do you add + C for a definite integral?
3. $\int_1^2 2x\,dx$?

Answers (click to reveal)

1. 4
2. no
3. $[x^2]_1^2 = 3$

Self-assess:

- need more help
- nearly there
- confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:
Definite integrals & area.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 43

End of Topic Test — Calculus. Revise Lessons 34–42.